Date: 05/05/2026 | MONTHLY CHALLENGE | SET B

Duration: 2 hours | III YEAR – SEMESTER 6

Question 1:

1. Problem Title & Link

Title: Maximum Path Sum in Binary Tree with Constraint

2. Problem Statement

You are given the root of a binary tree where each node contains an integer value (can be negative).

A **path** is defined as:

- Any sequence of nodes connected via parent-child edges
- The path **does not need to pass through root**
- The path **cannot reuse nodes**

- You are allowed to choose a path such that:

- You can include at most ONE negative node in the path

Return the **maximum sum of such a valid path**.

3. Examples

Example 1

Input:

```
  1
 / \
2   3
```

Output:

6

4. Constraints

- 1 <= nodes <= 10^5
- -1000 <= node.val <= 1000
- Must be efficient → **O(n)**

Question 2:

1. Problem Title & Link

Title: Word Ladder with Path Count

2. Problem Statement

You are given:

- A beginWord
- An endWord
- A list of words wordList

You can transform a word by:

- Changing **exactly one character at a time**
- Each transformed word must exist in wordList

Tasks

1. Return the **length of the shortest transformation sequence**
2. Return the **number of distinct shortest transformation sequences**

If no transformation exists:

Return: -1, 0

3. Examples

Example 1

Input:

beginWord = "hit"

endWord = "cog"

wordList = ["hot","dot","dog","lot","log","cog"]

Output:

length = 5

count = 2

4. Constraints

- 1 <= len(wordList) <= 5000
- Word length ≤ 10
- All words are lowercase
- All words have same length